Cambridge Assessment International Education

Science

Stage 4

Paper 2

2023

Cambridge Primary Progression Test

Mark Scheme

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General guidelines on marking

Many descriptive answers can be expressed in a variety of ways. Professional judgement can be used in these cases, providing it matches the marking points and further information in the mark scheme.

Answers may have words spelt incorrectly. Credit is normally given for phonetically correct answers, unless the word has a scientifically different meaning. For example, where the answer should be antennae, credit will be given for antena but not for anthen (too close to anther).

Only the science is being assessed, so answers do not need to be grammatically correct. Significant figures will be indicated in the question or in the mark scheme.

Unless specified all marking points are independent.

Annotations and abbreviations

/ **or**	alternate responses for the same marking point
( ) brackets	the words or units in brackets do not need to be stated, for example, (recycles or releases or provides) minerals = minerals scores the mark
<u>Underline</u>	exact word is required
Accept	an acceptable response
Do not accept	indicates an incorrect response that would contradict another otherwise correct alternative
Ignore	indicates an irrelevant answer that is not creditworthy. Full marks can still be achieved even with answers that are ignored
Note	provides extra information when necessary
ecf	error carried forward; marks are awarded if an incorrect response has been carried forward from earlier working, provided the subsequent working is correct
ora	or reverse argument; for example, as mass increases, volume increases could be written as mass decreases, volume decreases

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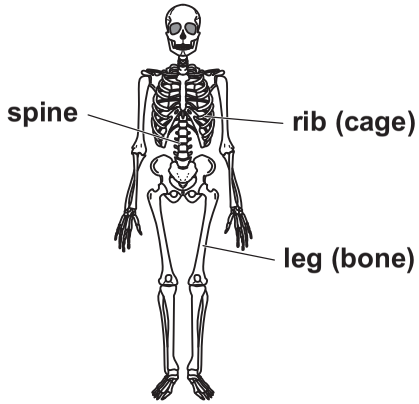
Question	Answer	Marks	Further Information
1(a)		**2**	**all three** correct = 2 marks **one** or **two** correct = 1 mark **Accept** vertebrae for spine **Accept** thigh (bone) **or** femur for leg (bone)
1(b)	one muscle contracts and the other muscle relaxes	**1**	more than **one** ticked = 0 marks **Accept** any indication of the correct answer, e.g. circling or underlining, but ticking takes precedence
1(c)	stronger bones **or** muscles	**1**	**Accept** any suitable answer, e.g. to keep fit **or** lose weight **or not** get overweight **Ignore** stay healthy **or** healthy bones **or** muscles **Ignore** stronger **or** healthy heart

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Question	Answer	Marks	Further Information
2(a)	magma chamber crater	**2**	each correct answer = 1 mark **Accept** magma **or** chamber
2(b)	earthquake	**1**	

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Question	Answer	Marks	Further Information
3(a)	**property of a liquid** / **explanation** liquids take the shape of the container they are in — particles move around in all directions liquids and solids **cannot** be squashed — particles are close together solids have a fixed shape — particles only move by vibrating particles move quickly particles are far apart	**3**	each correct answer = 1 mark if **two** lines from **one** property and **one** is incorrect = 0 marks for that property
3(b)	grains of sugar	**1**	more than **one** answer circled = 0 marks **Accept** any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence

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Question	Answer	Marks	Further Information
4(a)	switch is off **or** open	**1**	**Accept** break in circuit
4(b)	add another cell	**1**	**Accept** remove the switch **Accept** add another battery **Ignore** add a bigger battery
4(c)	reflects	**1**	**Do not accept** bounces **or** refracts

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th><th>Further Information</th></tr>
<tr><td>5(a)</td><td>any one from
same sized ice cubes
one ice cube used each time</td><td>1</td><td>Accept same number of ice cubes used each time

Ignore same equipment

Do not accept time taken to melt</td></tr>
<tr><td>5(b)</td><td><table>
<tr><th>(method to melt ice cube)</th><th>(time taken in) minutes/min</th></tr>
<tr><td>(beaker)</td><td>6</td></tr>
<tr><td>(hand)</td><td>3</td></tr>
<tr><td>(salt)</td><td>1</td></tr>
<tr><td>(in a beaker with) water</td><td>(2)</td></tr>
</table></td><td>2</td><td>water and all three correct numbers = 1 mark

correct units = 1 mark

but

Do not accept units in body of table for units mark</td></tr>
<tr><td>5(c)</td><td>(add) salt</td><td>1</td><td></td></tr>
<tr><td>5(d)</td><td>solidification</td><td>1</td><td>more than one circled = 0 marks

Accept any indication of the correct answer, e.g. ticking or underlining, but circling takes precedence</td></tr>
</table>

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Question	Answer	Marks	Further Information
6	leaf caterpillar chameleon carnivore	**3**	**all four** correct = 3 marks **two** or **three** correct = 2 marks **one** correct = 1 mark **Accept** consumer for carnivore

Question	Answer	Marks	Further Information
7(a)	ibis	**1**	
7(b)	(great egret) has legs longer than its body	**1**	**Note** assume answer refers to great egret unless otherwise specified **Accept ora** if snowy egret is specified
7(c)	spoonbill	**1**	
7(d)	uses its beak to scoop up large fish into its mouth	**1**	more than **one** ticked = 0 marks **Accept** any indication of the correct answer, e.g. circled or underlining, but ticking takes precedence

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Question	Answer	Marks	Further Information
8(a)	sound	**1**	**Ignore** noise
8(b)	heat	**1**	
8(c)	made destroyed	**2**	**both** needed in either order for the mark

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